Quantitative robust estimation for Gaussian inverse problems

A proof of the spectral conjecture and an exact boundary for isotropic input jittering

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Abstract

Krainovic, Soltanolkotabi and Heckel proposed a spectral estimator for Gaussian linear inverse problems by interchanging expectation and minimisation in a scalar trust-region dual. We prove its asymptotic optimality among all linear estimators, with a finite-dimensional additive robust-risk bound uniform over every forward operator, aspect ratio, measurement-noise level and positive adversarial radius. Normalised by signal energy, the regret is at most

$$\min\left\{1, \frac{2^{7/4}}{3^{3/4}} d^{-1/4} + \sqrt{2}\, d^{-1/2}\right\}.$$

The proof combines exact symmetry reduction, gain clipping and a Gaussian quadratic-process estimate controlling the singular dual boundary. An invariant version of the dual-gap estimate allows arbitrary Gaussian residual covariance. When the minimising dual parameter is positive, an explicit anisotropic training-noise covariance implements the spectral rule through ordinary population least squares. Under positive noise and active estimation, finite isotropic jittering implements that rule precisely when the positive singular spectrum is flat. Fixed unequal spectral bands yield a strictly positive limiting true-risk gap. A two-band example has a short analytic certificate giving a limiting gap greater than 0.141382 at unit signal energy, and a true finite-dimensional gap greater than 0.007292 at $d = 20000$. The source's estimator formula and qualitative spectral comparison are credited explicitly; the contributions here are quantitative proofs, precise conditions and exact certificates.

1 Problem, source and scope

Let d, m, n be positive integers with $n \geq d$. Let $U \in \mathbb{R}^{n\times d}$ satisfy $U^T U = I_d$, and let $A \in \mathbb{R}^{m\times n}$ be arbitrary. Consider

$$x = Uc, \qquad c \sim \mathcal{N}(0, aI_d), \qquad z \sim \mathcal{N}(0, bI_m), \qquad a = \frac{E}{d} > 0, \qquad b = \frac{\sigma_z^2}{m} \geq 0, \tag{1}$$

where c and z are independent and $E = \sigma_c^2 = \mathbb{E}\|x\|^2$ is the signal energy. For $\varepsilon > 0$ and $H \in \mathbb{R}^{n\times m}$, define

$$\mathcal{R}_\varepsilon(H) = \mathbb{E}\left[\max_{\|e\|\leq\varepsilon} \|H(Ax + z + e) - x\|^2\right]. \tag{2}$$

All vector norms are Euclidean. The adversarial maximum is inside the expectation: every realised sample has a separate perturbation budget.

Krainovic, Soltanolkotabi and Heckel [1, Conjecture 3.3] proposed an asymptotically optimal spectral estimator for (2), obtained by taking expectation before minimising a scalar trust-region dual. Their paper already supplies the spectral formula, dual representation and proposed interchange. The theorem below gives a uniform quantitative justification.

The conclusion compares linear estimators only. It requires no condition-number bound, limiting spectral distribution, fixed aspect ratio, positive measurement-noise assumption or positive lower bound on the nonzero singular values. The radius and noise level may depend on dimension. The conclusion is convergence of additive risk divided by signal energy, not a risk-ratio assertion or convergence of estimator matrices.

1.1 Relation to other robust-estimation problems

Zhang, Lee, Hassani and Matni [2] study per-realisation norm-bounded attacks in robust state estimation and linear inverse problems. Their inner attack is also a trust-region problem. Their Section 3.1 bounds the difference between adversarial and standard risks for a fixed estimator. The estimate proved here instead bounds the gap between an expected samplewise dual minimum and the minimum of its expectation, and then uses that estimate to justify a particular spectral estimator. The distinction concerns the quantity controlled, not the existence of a scalar dual.

Zhang, Blanchet, Marzouk, Nguyen and Wang [3] study Bayesian inverse estimation when an adversary changes a distribution within a Wasserstein neighbourhood of a nominal model. They establish strong duality and analyse a Gaussian least-favourable model and a Nash equilibrium. A Wasserstein transport budget constrains distributional transport cost, whereas (2) gives every realised measurement its own norm bound. Their minimax theorem does not itself establish the expectation–infimum estimate required here. Lotidis, Bambos, Blanchet and Li [4] also use Wasserstein model uncertainty, with martingale constraints for dynamical estimation.

Input jittering in this paper is a population linear-estimation construction. Iteration-wise noise injection in nonlinear reconstruction, such as Guan and Davenport [5], is a different procedure. No finite-sample neural-network optimisation guarantee follows from the results below.

2 Spectral estimator and main theorem

Choose a singular-value decomposition

$$AU = WDV^T,$$

where $W \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{d\times d}$ are orthogonal, and the rectangular diagonal matrix D has positive entries $s_1, \dots, s_r$, with $r = \operatorname{rank}(AU)$. Pad the list by $s_i = 0$ for $r < i \le d$, and put $t = b/a$.

For $\lambda \ge 0$, set

$$C_i(\lambda) = 1 + \lambda(s_i^2 + t), \qquad h_i(\lambda) = \begin{cases} \dfrac{2\lambda s_i}{C_i(\lambda) + \sqrt{C_i(\lambda)^2 - 4\lambda s_i^2}}, & s_i > 0, \\ 0, & s_i = 0. \end{cases} \tag{3}$$

In particular $h_i(0) = 0$. The discriminant is nonnegative since

$$C_i(\lambda)^2 - 4\lambda s_i^2 = [1 - \lambda(s_i^2 + t)]^2 + 4\lambda t.$$

The rationalised root avoids subtraction of nearly equal numbers. Define

$$\Phi(\lambda) = \lambda\varepsilon^2 + \frac{a}{2}\sum_{i=1}^{d}\left[1 - \lambda(s_i^2 + t) + \sqrt{[1 + \lambda(s_i^2 + t)]^2 - 4\lambda s_i^2}\right]. \tag{4}$$

Choose any minimiser $\lambda_* \geq 0$ of Φ. Let $K_* \in \mathbb{R}^{d\times m}$ have diagonal entries $h_i(\lambda_*)$, $i \leq r$, and every other entry zero. Set

$$H_* = UVK_*W^T. \tag{5}$$

Theorem 1 (uniform asymptotic optimality). The objective Φ is continuous and convex and has a minimiser in $[0, E/\varepsilon^2]$. Define

$$\eta_d = \frac{2^{7/4}}{3^{3/4}} d^{-1/4} + \sqrt{2}\, d^{-1/2}. \tag{6}$$

Then

$$0 \leq \mathcal{R}_\varepsilon(H_*) - \inf_H \mathcal{R}_\varepsilon(H) \leq E \min\{1, \eta_d\}. \tag{7}$$

Writing $V = \min_{\lambda\geq 0} \Phi(\lambda)$, the untruncated value comparison is

$$V - E\eta_d \leq \inf_H \mathcal{R}_\varepsilon(H) \leq \mathcal{R}_\varepsilon(H_*) \leq V. \tag{8}$$

Since $V \leq E$ and all risks are nonnegative, $E\eta_d$ in the lower bound can also be replaced by $E\min\{1, \eta_d\}$. Thus, for arbitrary sequences of operators, dimensions, noise levels and positive radii, the regret divided by E tends to zero as $d \to \infty$.

The simpler leading constant $2^{5/4}$ follows from the same proof. We give both calculations below. Neither the rate exponent nor the leading constant is asserted to be optimal.

3 Exact symmetry reduction and clipping

3.1 Existence and diagonal reduction

Projecting the output of H onto range(U) cannot increase the error for any x, z, e. It is therefore enough to take $H = UL$. Orthogonal changes of signal and measurement coordinates by V and W reduce the problem to estimating $c \sim \mathcal{N}(0, aI_d)$ from $Dc + z$ by a matrix $K \in \mathbb{R}^{d\times m}$, with the same isotropic noise and adversarial ball.

The risk is convex in K: each fixed-sample, fixed-perturbation loss is a convex quadratic, and a maximum followed by an expectation preserves convexity. Let T and S be diagonal sign matrices in signal and measurement space satisfying $SD = DT$. Use the same sign on the two coordinates corresponding to each positive singular value and independent signs on every remaining coordinate. Isotropy and invariance of the ball imply

$$\mathcal{R}_\varepsilon(TKS) = \mathcal{R}_\varepsilon(K).$$

Averaging K over this finite group removes every entry except K_{ii} for $i \leq r$. Convexity shows that the averaged diagonal estimator has no larger risk.

The risk is continuous in each fixed finite-dimensional matrix space. On a bounded set of matrices, dominated convergence applies with a constant multiple of $\|c\|^2 + \|z\|^2 + \varepsilon^2$ as a bound. Choose a deterministic e_0 such that $\|e_0\| = \varepsilon$ and $\|Ke_0\| = \varepsilon\|K\|_{\mathrm{op}}$. Averaging the losses at e_0 and $-e_0$ gives

$$\mathcal{R}_\varepsilon(K) \geq \mathbb{E}\|K(Dc + z) - c\|^2 + \varepsilon^2\|K\|_{\mathrm{op}}^2. \tag{9}$$

Thus sublevel sets are compact and a minimiser exists. A diagonal minimiser may be chosen. The zero estimator has risk $ad = E$.

For a diagonal estimator with gains h_i, the residual coordinates are independent centred Gaussians with variances

$$q_i = a(1 - s_i h_i)^2 + b h_i^2 \quad (i \le r), \qquad q_i = a \quad (i > r). \tag{10}$$

Pad $h_i = 0$ for $i > r$. This padding introduces no effective adversarial directions, including when $m < d$.

3.2 Clipping and residual dispersion

For $s_i > 0$, clip the gain to

$$0 \le h_i \le h_i^0, \qquad h_i^0 = \frac{a s_i}{a s_i^2 + b} = \frac{s_i}{s_i^2 + t}. \tag{11}$$

Clipping decreases both $|h_i|$ and q_i. The variance is a convex quadratic minimised at h_i^0, its value at zero is a, and a negative gain has variance at least a.

To justify risk monotonicity, temporarily regard the gains h and variances q as independent inputs. For fixed h, the function

$$v \longmapsto \max_{\|e\| \le \varepsilon} \|v + \operatorname{diag}(h) e\|^2$$

is convex. Increasing the Gaussian variances coordinatewise is equivalent to adding an independent centred Gaussian vector. Conditional Jensen's inequality shows that this cannot decrease expected loss. For fixed v, reducing $|h_i|$ coordinatewise shrinks the perturbation ellipsoid, so cannot increase its maximum loss. Applying the two facts successively proves that clipping cannot increase risk.

For a clipped diagonal robust minimiser, set

$$B = \max_i |h_i|, \qquad Q_0 = \sum_i q_i, \qquad C = \left(2 \sum_i q_i^2 \right)^{1/2}.$$

Then $0 \le q_i \le a$ and (9) gives

$$Q_0 + \varepsilon^2 B^2 \le E, \qquad \sum_i q_i^2 \le a Q_0, \qquad C \le \sqrt{\frac{2}{d}}\, E. \tag{12}$$

This is the uniform residual-dispersion bound. It does not depend on the singular values or on the choice of the robust minimiser.

4 Scalar dual and uniform expectation–infimum estimate

4.1 Trust-region identity, including singular boundaries

For $v, h \in \mathbb{R}^d$ and $B = \max_i |h_i|$, the trust-region identity is

$$\max_{\|e\| \le \varepsilon} \|v + \operatorname{diag}(h) e\|^2 = \inf_{\lambda > B^2} \left[\lambda \varepsilon^2 + \sum_i v_i^2 \frac{\lambda}{\lambda - h_i^2} \right]. \tag{13}$$

Square completion gives the upper bound for every $\lambda > B^2$. The stationary attack is

$$e_i(\lambda) = \frac{h_i v_i}{\lambda - h_i^2}.$$

Its norm is continuous and nonincreasing in λ and tends to zero at infinity. If its norm reaches ε at a strictly feasible λ, substitution attains equality. Otherwise let $\lambda \downarrow B^2$. When $B > 0$, finiteness of the limiting norm forces $h_i v_i = 0$ in the top eigenspace. Any missing attack norm can then be added in that eigenspace, giving the limiting value. This is the usual hard

case. If $B = 0$, both sides of (13) equal $\|v\|^2$. The strict-domain infimum therefore handles all singular boundaries without an inverse of a singular matrix.

For independent standard Gaussian g_i, set $v_i = \sqrt{q_i} g_i$ and define

$$\begin{aligned} R(h,q) &= \mathbb{E} \inf_{\lambda > B^2} \left[\lambda \varepsilon^2 + \sum_i q_i g_i^2 \frac{\lambda}{\lambda - h_i^2} \right], \\ S(h,q) &= \inf_{\lambda > B^2} \left[\lambda \varepsilon^2 + \sum_i q_i \frac{\lambda}{\lambda - h_i^2} \right]. \end{aligned} \tag{14}$$

Choosing any fixed deterministic dual parameter before expectation shows $R(h,q) \le S(h,q)$.

Lemma 2 (uniform dual gap). For arbitrary $q_i \ge 0$, $h_i \in \mathbb{R}$, $B = \max_i |h_i|$ and $C = (2\sum_i q_i^2)^{1/2}$, every $\delta > 0$ satisfies

$$0 \le S(h,q) - R(h,q) \le \varepsilon^2 \delta + C \left(1 + \frac{B^2}{\delta} \right). \tag{15}$$

Proof. Write

$$F_g(\lambda) = \lambda \varepsilon^2 + \sum_i q_i g_i^2 \frac{\lambda}{\lambda - h_i^2},$$

and let R_δ, S_δ be obtained by restricting the corresponding infima to $\lambda \ge B^2 + \delta$. Each rational summand is decreasing in λ. Replacing λ by $\max\{\lambda, B^2 + \delta\}$ therefore increases F_g by at most $\varepsilon^2 \delta$. Hence

$$R \le R_\delta \le R + \varepsilon^2 \delta, \qquad S \le S_\delta. \tag{16}$$

The centred process

$$X(\lambda) = \sum_i q_i (g_i^2 - 1) \frac{\lambda}{\lambda - h_i^2}$$

converges at infinity to $X(\infty) = \sum_i q_i (g_i^2 - 1)$, with $\mathbb{E}|X(\infty)| \le C$. Independence and differentiation give

$$\begin{aligned} X'(\lambda) &= -\sum_i q_i (g_i^2 - 1) \frac{h_i^2}{(\lambda - h_i^2)^2}, \\ \mathbb{E}|X'(\lambda)| &\le \left[2 \sum_i q_i^2 \frac{h_i^4}{(\lambda - h_i^2)^4} \right]^{1/2} \le \frac{C B^2}{(\lambda - B^2)^2}. \end{aligned}$$

The fundamental theorem of calculus and Tonelli's theorem imply

$$\begin{aligned} \mathbb{E} \sup_{\lambda \ge B^2 + \delta} |X(\lambda)| &\le \mathbb{E}|X(\infty)| + \int_{B^2 + \delta}^{\infty} \mathbb{E}|X'(u)| \, du \\ &\le C \left(1 + \frac{B^2}{\delta} \right). \end{aligned} \tag{17}$$

Since $\inf(F + X) \ge \inf F - \sup |X|$, one has $S_\delta - R_\delta \le \mathbb{E} \sup |X|$. Combining this with (16) proves (15). All terms are finite because the cutoff is positive. □

The lemma does not require any relationship between q and h. Its argument also works for independent nonnegative unit-mean multipliers other than g_i^2, with C^2 replaced by the sum of their weighted variances. Nonnegativity is needed for the cutoff step. The exact symmetry reduction above still relies on Gaussian rotational invariance.

5 Surrogate minimisation and completion of the main proof

5.1 One coordinate at fixed dual parameter

For $s_i > 0$ and $\lambda > 0$, minimise, or take the boundary infimum of,

$$\frac{\lambda[a(1 - s_i h)^2 + bh^2]}{\lambda - h^2}, \qquad |h| < \sqrt{\lambda}. \tag{18}$$

The stationary equation is

$$s_i h^2 - [1 + \lambda(s_i^2 + t)]h + \lambda s_i = 0. \tag{19}$$

The smaller root is (3). It belongs to

$$0 \le h_i(\lambda) \le \min\left\{\sqrt{\lambda}, \frac{s_i}{s_i^2 + t}\right\},$$

and the coordinate infimum is

$$\frac{a}{2}\left[1 - \lambda(s_i^2 + t) + \sqrt{[1 + \lambda(s_i^2 + t)]^2 - 4\lambda s_i^2}\right]. \tag{20}$$

For $b > 0$, the minimum is attained in the strict domain. For $b = 0$, (20) reduces to $a(1 - \lambda s_i^2)_+$ and

$$h_i(\lambda) = \min\{\lambda s_i, 1/s_i\}. \tag{21}$$

At $\lambda s_i^2 = 1$, the boundary $h_i^2 = \lambda$ has zero residual variance; it is handled by the limiting dual. For $s_i = 0$, the best gain is zero and its contribution is a.

Consequently

$$\inf_h S(h, q(h)) = \inf_{\lambda \ge 0} \Phi(\lambda). \tag{22}$$

These are infima over the same feasible pairs (h, λ), not a minimax interchange. The stated boundary conventions extend the identity to zero noise and $\lambda = 0$.

Every summand of Φ is convex. For $t > 0$ this follows by differentiating its square root and using $(s_i^2 + t)^2 - (t - s_i^2)^2 = 4s_i^2 t \ge 0$. For $t = 0$, it is a positive-part affine function. Since $\Phi(0) = E$ and each coordinate contribution is nonnegative, Φ is continuous and coercive for $\varepsilon > 0$ and has a minimiser in $[0, E/\varepsilon^2]$. The gains at any minimising λ_* attain the minimum surrogate, including $\lambda_* = 0$ and the noiseless boundary cases.

The translation to [1] is $s_i = \lambda_i$, $h_i = \sigma_i$, and $t = (d/m)\sigma_z^2/\sigma_c^2$. The scalar objective and gains agree algebraically with the source conjecture.

5.2 Regret and the sharpened coefficient

Let h_{opt} be a clipped diagonal robust minimiser and h_* a surrogate minimiser. Then

$$\begin{aligned} R(h_*) - R(h_{\mathrm{opt}}) &\le S(h_*) - R(h_{\mathrm{opt}}) \\ &\le S(h_{\mathrm{opt}}) - R(h_{\mathrm{opt}}). \end{aligned} \tag{23}$$

Apply Lemma 2 to h_{opt}. The direct inequalities $C \le \sqrt{2}E/\sqrt{d}$ and $B^2 \le E/\varepsilon^2$, together with $\delta = 2^{1/4}E/(\varepsilon^2 d^{1/4})$, give

$$R(h_*) - R(h_{\mathrm{opt}}) \le E\left(2^{5/4} d^{-1/4} + \sqrt{2} d^{-1/2}\right). \tag{24}$$

For the improved coefficient, retain the joint constraint in (12):

$$
\begin{aligned}
CB^2 &\le \frac{\sqrt{2aQ_0}(E-Q_0)}{\varepsilon^2} \\
&\le \frac{2\sqrt{2}}{3\sqrt{3}}\,\frac{E^2}{\varepsilon^2\sqrt{d}}.
\end{aligned}
\tag{25}
$$

The last inequality follows from

$$
\max_{0\le Q_0\le E}(E-Q_0)\sqrt{Q_0} = \frac{2E^{3/2}}{3\sqrt{3}},
$$

attained at $Q_0 = E/3$, and $a = E/d$. Optimising the cutoff terms gives

$$
\inf_{\delta>0}\left(\varepsilon^2\delta + \frac{CB^2}{\delta}\right) = 2\varepsilon\sqrt{CB^2} \le \frac{2^{7/4}}{3^{3/4}}Ed^{-1/4}. \tag{26}
$$

The equality is an infimum with $\delta \downarrow 0$ if $CB^2 = 0$. Adding $C \le \sqrt{2}Ed^{-1/2}$ proves the untruncated error bound $E\eta_d$.

For the value comparison, write $V = \min\Phi$. The same calculation yields

$$
V \le S(h_{\mathrm{opt}}) \le R(h_{\mathrm{opt}}) + E\eta_d.
$$

Together with $R(h_*) \le S(h_*) = V$, this proves (8) first, with the untruncated error. Since $V \le E$ and $R(h_{\mathrm{opt}}) \ge 0$, also $V - R(h_{\mathrm{opt}}) \le E$. The lower value bound can therefore use $E\min\{1,\eta_d\}$. Finally $R(h_*) \le V \le E$ and nonnegativity truncate the regret in the same way. This proves Theorem 1. □

The estimate was applied only to an arbitrary clipped robust optimiser, with constants independent of that optimiser. No unproved uniform convergence over estimators is assumed.

6 Invariant extension to correlated Gaussian residuals

Proposition 3 (arbitrary Gaussian covariance). Let $v \sim \mathcal{N}(0,Q)$ in $\mathbb{R}^k$, with any positive-semidefinite covariance Q, and let $H \in \mathbb{R}^{k\times m}$ be arbitrary. Set $B = \|H\|_{\mathrm{op}}$ and

$$
\begin{aligned}
R &= \mathbb{E}\max_{\|e\|\le\varepsilon}\|v+He\|^2, \\
S &= \inf_{\lambda>B^2}\left[\lambda\varepsilon^2 + \operatorname{tr}\Big(Q(I-HH^T/\lambda)^{-1}\Big)\right].
\end{aligned}
$$

For every $\delta > 0$,

$$
0 \le S - R \le \varepsilon^2\delta + \sqrt{2}\|Q\|_F\left(1 + \frac{B^2}{\delta}\right). \tag{27}
$$

No commutation between Q and HH^T is assumed.

Proof. The trust-region dual and cutoff argument are unchanged. Put

$$
M_\lambda = (I - HH^T/\lambda)^{-1}, \qquad X(\lambda) = v^T M_\lambda v - \operatorname{tr}(QM_\lambda).
$$

For symmetric M, the Gaussian quadratic-form identity gives $\operatorname{Var}(v^TMv) = 2\operatorname{tr}(QMQM)$. Moreover,

$$
\operatorname{tr}(QMQM) \le \|M\|_{\mathrm{op}}^2\operatorname{tr}(Q^2). \tag{28}
$$

Indeed, diagonalising Q with eigenvalues $q_i \ge 0$ gives

$$\begin{aligned}\operatorname{tr}(QMQM) &= \sum_{i,j} q_i q_j M_{ij}^2 \\ &\le \frac{1}{2}\sum_{i,j}(q_i^2+q_j^2)M_{ij}^2 \\ &= \sum_i q_i^2 (M^2)_{ii} \le \|M\|_{\mathrm{op}}^2 \sum_i q_i^2.\end{aligned}$$

At infinity, $X(\infty) = \|v\|^2 - \operatorname{tr} Q$, so $\mathbb{E}|X(\infty)| \le \sqrt{2}\|Q\|_F$. Also

$$M_\lambda' = -HH^T(\lambda I - HH^T)^{-2}, \qquad \|M_\lambda'\|_{\mathrm{op}} \le \frac{B^2}{(\lambda - B^2)^2}.$$

Using (28), the same integral argument as (17) proves (27). □

For $Q \neq 0$, define

$$d_{\mathrm{eff}} = \frac{(\operatorname{tr} Q)^2}{\operatorname{tr}(Q^2)}.$$

Then $\|Q\|_F = \operatorname{tr} Q/\sqrt{d_{\mathrm{eff}}}$. The covariance bound reflects dispersion of variance; adding zero-variance coordinates does not improve this scale. The main inverse-problem theorem uses clipping to obtain the required dispersion uniformly over its unknown robust optimiser.

7 Constructive anisotropic input jittering

Assume $\lambda_* > 0$. In the measurement singular-vector basis W, choose

$$w_i = \frac{as_i}{h_i(\lambda_*)} - (as_i^2 + b) = \frac{a[1 - s_i h_i(\lambda_*)]}{\lambda_*} \ge 0, \qquad i \le r. \tag{29}$$

For $r < i \le m$, set $w_i = a/\lambda_* > 0$. Let ξ be centred Gaussian, independent of (c, z), with covariance

$$\operatorname{Cov}(\xi) = W \operatorname{diag}(w_1, \dots, w_m) W^T. \tag{30}$$

Consider ordinary population least-squares training,

$$\mathcal{J}(H) = \mathbb{E}\|H(Ax + z + \xi) - x\|^2. \tag{31}$$

Corollary 4 (exact population implementation). The unique linear minimiser of (31) is H_* and therefore satisfies Theorem 1.

Proof. In singular coordinates the linear minimum mean-square-error gains are

$$\frac{as_i}{as_i^2 + b + w_i}, \qquad i \le r,$$

with every other entry zero. Formula (29) makes them exactly $h_i(\lambda_*)$. Dividing (19) by $\lambda_* h_i$ and multiplying by a gives the second expression in (29). Since $0 \le s_i h_i \le 1$, the variances are nonnegative. The input covariance is positive on informative directions because $as_i^2 > 0$, and on all remaining measurement directions because the chosen $w_i > 0$. Hence the least-squares minimiser is unique. □

The covariance is directly computable. Anisotropic Gaussian noise and its Wiener-filter algebra are not claimed to be new concepts; the result pairs this particular covariance with the proved robust-risk approximation.

If $\lambda_* = 0$, prescribe $H_* = 0$ directly. A finite jitter covariance cannot produce zero gains on every informative direction, so that endpoint is excluded from Corollary 4. When $A = 0$, the

zero estimator is already available. When $b = 0$ and $h_i = 1/s_i$, its required jitter variance is zero, which is valid.

8 Exact isotropic implementation and its boundary

Write $\kappa = w/a$ for a common isotropic jitter variance relative to the signal-coordinate variance. Its gains are

$$h_i^{\mathrm{iso}}(\kappa) = \frac{s_i}{s_i^2 + t + \kappa}, \qquad \kappa \geq 0. \tag{32}$$

The source already identifies equal singular values as an optimality case and discusses a typically small gap for unequal singular values [1, Section 3.3]. The next results turn that comparison into an exact classification under positive noise and a strict limiting gap for the true samplewise robust risk.

Proposition 5 (flat-spectrum characterisation). Suppose $t > 0$, $\lambda_* > 0$, and AU has at least one positive singular value. A finite isotropic jitter variance produces exactly H_* if and only if all positive singular values of AU are equal.

Proof. If the isotropic gains equal the proposed robust gains, (29) gives

$$\kappa = \frac{1 - s_i h_i^{\mathrm{iso}}(\kappa)}{\lambda_*} = \frac{t + \kappa}{\lambda_*(s_i^2 + t + \kappa)}$$

for each $s_i > 0$. Equivalently,

$$\lambda_* \kappa (s_i^2 + t + \kappa) = t + \kappa. \tag{33}$$

Since $t > 0$, the ratio κ must be positive; (33) then forces equality of all positive s_i^2. Conversely, if these singular values agree, (29) gives the same positive ratio on every informative direction. Using it isotropically yields the same gains. Zero singular directions have zero reconstruction gain regardless. □

For squared singular value $u > 0$, the required ratio solves

$$\lambda_* \kappa^2 + [\lambda_*(u + t) - 1]\kappa - t = 0. \tag{34}$$

It has one positive root, which decreases strictly with u:

$$\frac{d\kappa}{du} = -\frac{\lambda_* \kappa}{2\lambda_* \kappa + \lambda_*(u + t) - 1} < 0.$$

The denominator is the positive square root of the quadratic discriminant. Thus the prescribed covariance injects more training noise into less informative directions.

Proposition 5 concerns exact equality of estimators. That alone does not prove a positive finite-dimensional risk gap. The next theorem supplies the additional limit argument.

9 Persistent asymptotic gap for fixed unequal bands

Normalise $E = 1$. Fix positive singular values $s_1, \ldots, s_J$, positive rational proportions p_j with $\sum_j p_j = 1$, and a positive noise ratio t. Assume

$$0 < \varepsilon^2 < \sum_j p_j s_j^2. \tag{35}$$

Let $d \to \infty$ through common multiples of the proportions' denominators. Take $m = n = d$, $U = I_d$, $c \sim \mathcal{N}(0, I_d/d)$ and $z \sim \mathcal{N}(0, tI_d/d)$. Let A_d be diagonal with s_j repeated $p_j d$ times. The operators are deterministic.

For a bandwise constant gain vector $h \in \mathbb{R}^J$, write

$$q_j(h) = (1 - s_j h_j)^2 + t h_j^2$$

and

$$S(h) = \inf_{\lambda > \max_j h_j^2} \left[\lambda \varepsilon^2 + \sum_j p_j q_j(h) \frac{\lambda}{\lambda - h_j^2} \right]. \tag{36}$$

The deterministic primal form is

$$S(h) = \max_{\sum_j e_j^2 \le \varepsilon^2} \sum_j \left[\sqrt{p_j q_j(h)} + h_j e_j \right]^2. \tag{37}$$

It follows from (13) using deterministic residual coordinates $\sqrt{p_j q_j(h)}$. Equivalently, in each equal-gain band the worst perturbation aligns with the band's residual, so only its norm matters.

Define

$$\begin{aligned} V_* &= \min_{\lambda \ge 0} \left\{ \lambda \varepsilon^2 + \frac{1}{2} \sum_j p_j \left[1 - \lambda(s_j^2 + t) + \sqrt{[1 + \lambda(s_j^2 + t)]^2 - 4\lambda s_j^2} \right] \right\}, \\ V_{\mathrm{iso}} &= \inf_{\kappa \ge 0} S(h^{\mathrm{iso}}(\kappa)), \qquad \Delta = V_{\mathrm{iso}} - V_*. \end{aligned} \tag{38}$$

Theorem 6 (strict limiting isotropic gap). If at least two positive bands have distinct singular values, then $\Delta > 0$ and

$$\lim_{d \to \infty} \left[\inf_{\kappa \ge 0} \mathcal{R}_{\varepsilon,d}(H_{\mathrm{iso}}(\kappa)) - \inf_H \mathcal{R}_{\varepsilon,d}(H) \right] = \Delta > 0. \tag{39}$$

The anisotropic-jitter estimator has risk converging to V_*. If all singular values in the fixed list agree, then $\Delta = 0$.

Proof. The scalar objective in (38) has derivative $\varepsilon^2 - \sum_j p_j s_j^2 < 0$ at zero. It therefore has a positive minimiser. The minimiser is unique because its second derivative is

$$\sum_j \frac{2 p_j s_j^2 t}{\left([1 + \lambda(s_j^2 + t)]^2 - 4\lambda s_j^2 \right)^{3/2}} > 0. \tag{40}$$

At $\lambda_* > 0$, every coordinate minimisation has a unique interior gain because $t > 0$.

The full surrogate has a unique gain minimiser. A minimising vector cannot be zero because $V_* < 1 = S(0)$. For a nonzero vector, every $q_j(h)$ is positive, so its scalar dual objective diverges at the largest gain pole and at infinity. Its dual optimum is attained in the strict domain. Equality with V_* forces that optimum to be λ_* and forces every coordinate gain to its unique minimiser. Thus h_* is the only minimising gain vector.

The isotropic path extends continuously to $\kappa = \infty$ by $h^{\mathrm{iso}}(\infty) = 0$. Formula (37) makes S continuous in h. Hence the isotropic objective is continuous on the compactification $[0, \infty]$ and attains its infimum. The endpoint at infinity has value $1 > V_*$. At finite κ, Proposition 5 excludes equality with h_* whenever two bands differ. Uniqueness of the global surrogate minimiser yields $V_{\mathrm{iso}} > V_*$.

For every κ, the isotropic gains are clipped and the dimension-d residual coordinate variances are at most $1/d$. Their operator norms obey the fixed bound

$$\overline{B} = \max_j \frac{s_j}{s_j^2 + t} < \infty.$$

Lemma 2 therefore gives, uniformly over $\kappa \ge 0$,

$$0 \le S(h^{\mathrm{iso}}(\kappa)) - \mathcal{R}_{\varepsilon,d}(H_{\mathrm{iso}}(\kappa)) \le \varepsilon^2\delta + \sqrt{2/d}\left(1 + \frac{\overline{B}^2}{\delta}\right). \tag{41}$$

Taking $\delta = d^{-1/4}$ makes the right side tend to zero. The best true isotropic risk tends to V_{iso}. For these repeated bands, the scalar objective of Theorem 1 is exactly (38), so (8) shows that the global optimal linear risk and the selected spectral risk both tend to V_*. Their difference gives (39). For a flat spectrum, one isotropic variance implements h_*, proving the converse. □

At general fixed signal energy E, the activity condition becomes $\varepsilon^2 < E\sum_j p_j s_j^2$. Risks are E times their normalised counterparts with radius squared ε^2/E. The result is an additive risk difference.

10 An analytic two-band separation certificate

Take

$$(s_1, s_2) = \left(\frac{2}{5}, 1\right), \qquad (p_1, p_2) = \left(\frac{1}{5}, \frac{4}{5}\right), \qquad t = \frac{1}{1000}, \qquad \varepsilon^2 = \frac{1}{5}, \qquad E = 1. \tag{42}$$

Condition (35) holds because $\sum_j p_j s_j^2 = 104/125$. Dimensions may be $5, 10, 15, \ldots$.

For orientation, numerical scalar optimisation gives $\lambda_* \simeq 1.0419687516$, spectral gains $(0.4162671156, 0.9827181294)$, required anisotropic ratios $(0.7999214491, 0.0165857858)$, and $V_* \simeq 0.3889178775$. The best isotropic ratio is approximately 0.3793690380, with $V_{\mathrm{iso}} \simeq 0.5312794237$ and gap approximately 0.1423615462. The proof below does not rely on these decimals, numerical optimisation or a grid.

Proposition 7 (analytic separation). For (42),

$$V_{\mathrm{iso}} \ge \frac{5303}{10000} = 0.5303, \qquad V_* < \frac{388918}{1000000} = 0.388918,$$

and consequently

$$\Delta > \frac{141382}{1000000} = 0.141382. \tag{43}$$

Proof: lower bound. Put $c = \kappa + t > 0$. The isotropic gains and variances satisfy

$$h_j = \frac{s_j}{s_j^2 + c}, \qquad q_j = \frac{c^2 + ts_j^2}{(s_j^2 + c)^2}, \qquad \sqrt{q_j} \ge \frac{c}{s_j^2 + c}. \tag{44}$$

Use the single fixed feasible attack

$$e = \frac{(4, 3)}{5\sqrt{5}}, \qquad e_1^2 + e_2^2 = \frac{1}{5}.$$

All entries are nonnegative. Substituting into (37), then using (44), gives

$$S(h^{\mathrm{iso}}(\kappa)) \ge L(c) := \frac{1}{5}\left[\left(\frac{c + 8/25}{c + 4/25}\right)^2 + \left(\frac{2c + 3/5}{c + 1}\right)^2\right]. \tag{45}$$

Set $y = 25c \ge 0$. Exact algebra gives

$$L(c) - \frac{5303}{10000} = \frac{F(y)}{10000(y + 4)^2(y + 25)^2}, \tag{46}$$

where

$$F(y) = 4697y^4 + 8426y^3 - 1004423y^2 + 1162600y + 34170000. \tag{47}$$

The following identity proves positivity without root approximation:

$$\begin{aligned} F(y) = (y-9)^2(4697y^2 + 92972y + 288616) \\ - 1173044(y-9) + 234708. \end{aligned} \tag{48}$$

For $0 \le y \le 9$, all terms on the right are nonnegative and the constant term is positive. For $y \ge 9$, write $z = y - 9 \ge 0$. The quadratic coefficient in parentheses is increasing for $y \ge 0$ and has value 1505821 at $y = 9$. Thus

$$F(y) \ge 1505821z^2 - 1173044z + 234708.$$

This quadratic has positive leading coefficient and discriminant

$$1173044^2 - 4 \cdot 1505821 \cdot 234708 = -37680715136 < 0. \tag{49}$$

Therefore $F(y) > 0$ for every $y \ge 0$. Equations (45)–(46) give the claimed uniform lower bound on V_{iso}.

Proof: upper bound. Choose the rational feasible dual and gain tuple

$$\lambda = \frac{104196875}{10^8}, \qquad h_1 = \frac{416267115648}{10^{12}}, \qquad h_2 = \frac{982718129443}{10^{12}}. \tag{50}$$

Both inequalities $\lambda > h_j^2$ hold. With $q_j = (1 - s_j h_j)^2 + t h_j^2$, the gap between 0.388918 and its feasible dual value is exactly

$$\begin{aligned} &\frac{388918}{10^6} - \left[\lambda\varepsilon^2 + \sum_j p_j q_j \frac{\lambda}{\lambda - h_j^2}\right] \\ &\quad = \frac{3168557554921612509053869041613787853379731}{25868592782876951709628705995150319490966201600000} > 0. \end{aligned} \tag{51}$$

Since V_* is the minimum over all feasible gains and dual parameters, (51) proves the strict upper bound. Subtracting proves (43). □

10.1 Explicit finite-dimensional separation

Corollary 8. For (42) and every positive dimension d divisible by five,

$$\begin{aligned} &\inf_{\kappa \ge 0} \mathcal{R}_{\varepsilon,d}(H_{\mathrm{iso}}(\kappa)) - \inf_H \mathcal{R}_{\varepsilon,d}(H) \\ &\qquad > \frac{141382}{1000000} - \left(\frac{128}{27d}\right)^{1/4} - \sqrt{\frac{2}{d}}. \end{aligned} \tag{52}$$

At $d = 20000$, and at every larger multiple of five, this true risk difference is strictly greater than 0.007292.

Proof. At each fixed dimension, the isotropic path compactifies continuously by the zero estimator at $\kappa = \infty$. The risk is continuous on that compactification by dominated convergence because gains and residual second moments are uniformly bounded. A best estimator on the compactification exists and its risk is at most one. The finite-κ infimum has the same value, since zero is a limit of finite-κ estimators.

At a minimising isotropic estimator, all coordinate variances are at most $1/d$, and (9) gives $Q_0 + \varepsilon^2 B^2 \le 1$. The sharpened calculation (25)–(26) therefore bounds its surrogate-minus-true-risk gap by

$$\eta_d = \left(\frac{128}{27d}\right)^{1/4} + \sqrt{\frac{2}{d}}.$$

Its surrogate value is at least V_{iso}, whereas the global optimal linear true risk is at most V_*. The difference is at least $\Delta - \eta_d$, and (43) makes (52) strict. This coercivity bound is applied at a best isotropic estimator, not asserted uniformly over every isotropic gain.

The numerical dimension threshold also follows from exact arithmetic:

$$\left(\frac{12409}{100000}\right)^4 - \frac{128}{27 \cdot 20000} = \frac{192972343173947}{270000000000000000000} > 0, \qquad \sqrt{\frac{2}{20000}} = \frac{1}{100}. \tag{53}$$

Hence $\eta_{20000} < 0.13409$ and $0.141382 - 0.13409 = 0.007292$. Since η_d decreases with d, the same lower bound holds for every larger allowed dimension. □

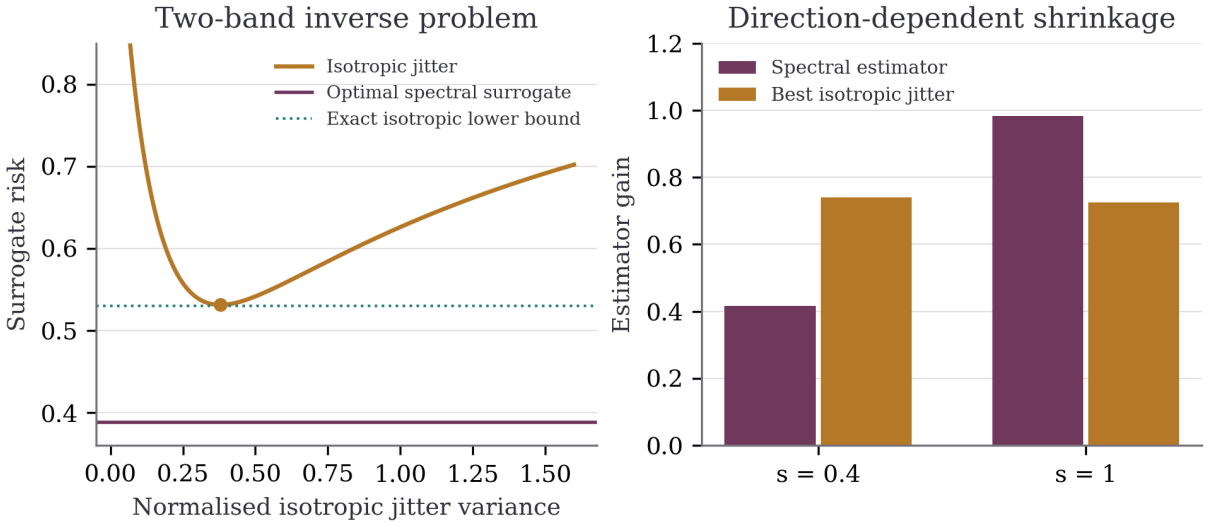


Figure 1: Numerical illustration of the two-band problem (42). The curves show the deterministic surrogate and the exact isotropic lower bound 0.5303; the bars compare the numerically minimising gains. The separation is proved by the rational certificate (44)–(51), independently of these plots or numerical optimisation.

11 Endpoint checks, reproducibility and limitations

At $A = 0$, the proposed gains are all zero and both true and surrogate risks equal E. For $A = I_n$, $b = 0$ and $\varepsilon < \sqrt{E}$, all d informative singular values equal one, so

$$\Phi(\lambda) = \lambda\varepsilon^2 + E(1-\lambda)_+.$$

Its minimum is at $\lambda = 1$ and gives $H_* = UU^T$, with robust risk exactly ε^2. This equals I_n only in the full-dimensional case. At $\varepsilon = 0$, the usual linear minimum mean-square-error estimator is exactly optimal; its informative gains arise as $\lambda \to \infty$. The positive-radius coercivity argument is not needed at that separate endpoint.

The deterministic surrogate need not be the exact finite-dimensional optimum. In a scalar problem with $a = s = 1$, $b = 0$ and $\sqrt{2/\pi} < \varepsilon < 1$, the prescribed gain is $h_* = 1$. For $0 \le h \le 1$, however,

$$R(h) = (1-h)^2 + \varepsilon^2 h^2 + 2\varepsilon\sqrt{2/\pi}\, h(1-h). \tag{54}$$

Its left derivative at $h = 1$ is positive, so a slightly smaller gain reduces true risk. At $\varepsilon = 0.9$, the prescribed risk is 0.81, while the scalar minimum is approximately 0.7874046283 at $h \simeq 0.7541413058$. This confirms the need for an asymptotic risk statement.

The exact symmetry, clipping and boundary arguments cover rectangular measurements, rank deficiency and zero measurement noise. In contrast, the isotropic classification and fixed-band gap use their stated additional hypotheses: positive noise ratio and positive dual minimiser, and for the gap theorem, positive fixed band proportions and the active-estimation condition (35). These conditions must not be dropped by importing the broader scope of Theorem 1.

The central proof is analytic. The polynomial identities and rational inequalities in the final certificate are displayed explicitly so that they can be checked with integer and rational arithmetic. Numerical optimisation is used only for the illustrative decimal values. Independent internal agents examined the proof, reproduced exact arithmetic and checked the statement against the originating source. These checks are not external peer review or proof-assistant formalisation.

The recorded prior-art search was last checked on 17 October 2026. It located the originating conjecture and adjacent work, but no later proof of the exact general statement in the searched corpus. This is bounded evidence, not proof of historical priority. The estimator formula, trust-region dual, isotropic Wiener gains and qualitative spectral limitation are credited to the source. The paper does not establish optimality over nonlinear estimators, an optimal convergence exponent or constant, a matrix-norm convergence rate, or practical neural-network training guarantees.

Claim-to-evidence map

Claim	Analytic support	Executable support	Not established by the check
Uniform additive regret bound	Theorem 1, Lemma 2, (23)–(26)	`inverse_risk_checks.py` finite cases	the uniform theorem itself
Arbitrary-covariance dual gap	Proposition 3	analytic proof; no packaged executable check	optimality of the constant
Anisotropic implementation	Corollary 4 and (29)	related scalar-root and spectral-gain checks only	the covariance identity or finite-sample guarantees
Isotropic boundary and persistent gap	Proposition 5 and Theorem 6	one-positive-band finite checks plus exact two-band certificate	the analytic general theorem or equality outside its hypotheses
Explicit two-band separation	Proposition 7 and Corollary 8	`analytic_isotropic_gap.py` exact arithmetic	optimality of the example or dimension threshold

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